

Introduction

CHAPTER 1

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- ❖ What is computer vision?
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What is computer vision?

Human Vision System

- ❖ It perceives the three-dimensional structure of the world around us with apparent ease.
- ❖ It has no problem interpreting the subtle variations
 - in translucency and shading in this photo and
 - correctly segmenting the object from its background.



Figure 1: Human visual system

What is computer vision?

- ❖ *Computer Vision* is the process whereby a machine, usually a digital computer, automatically processes an image and reports “what is in the image.”
- ❖ In *Computer Vision*, a camera (or several cameras) is linked to a computer.
- ❖ The computer interprets images of a real scene to obtain information useful for tasks such as navigation, manipulation and recognition.

What is computer vision?

Why study computer vision

- ❖ Images and movies have become ubiquitous in both production and consumption.
- ❖ Therefore, applications to manipulate images become core.
- ❖ Fast-growing collection of useful applications:
 - Surveillance
 - Building 3D representations
 - Face finding
- ❖ Greater understanding of human vision.

What is computer vision?

Goal of Computer Vision

- ❖ To bridge the gap between pixels and “meaning”.
- ❖ Goal of computer vision is to write the computer programs that can interpret images.



0	3	2	5	4	7	6	9	8
3	0	1	2	3	4	5	6	7
2	1	0	3	2	5	4	7	6
5	2	3	0	1	2	3	4	5
4	3	2	1	0	3	2	5	4
7	4	5	2	3	0	1	2	3
6	5	4	3	2	1	0	3	2
9	6	7	4	5	2	3	0	1
8	7	6	5	4	3	2	1	0

Figure 2: Machin Interpretation

What is computer vision?

Computer Vision vs Human Vision

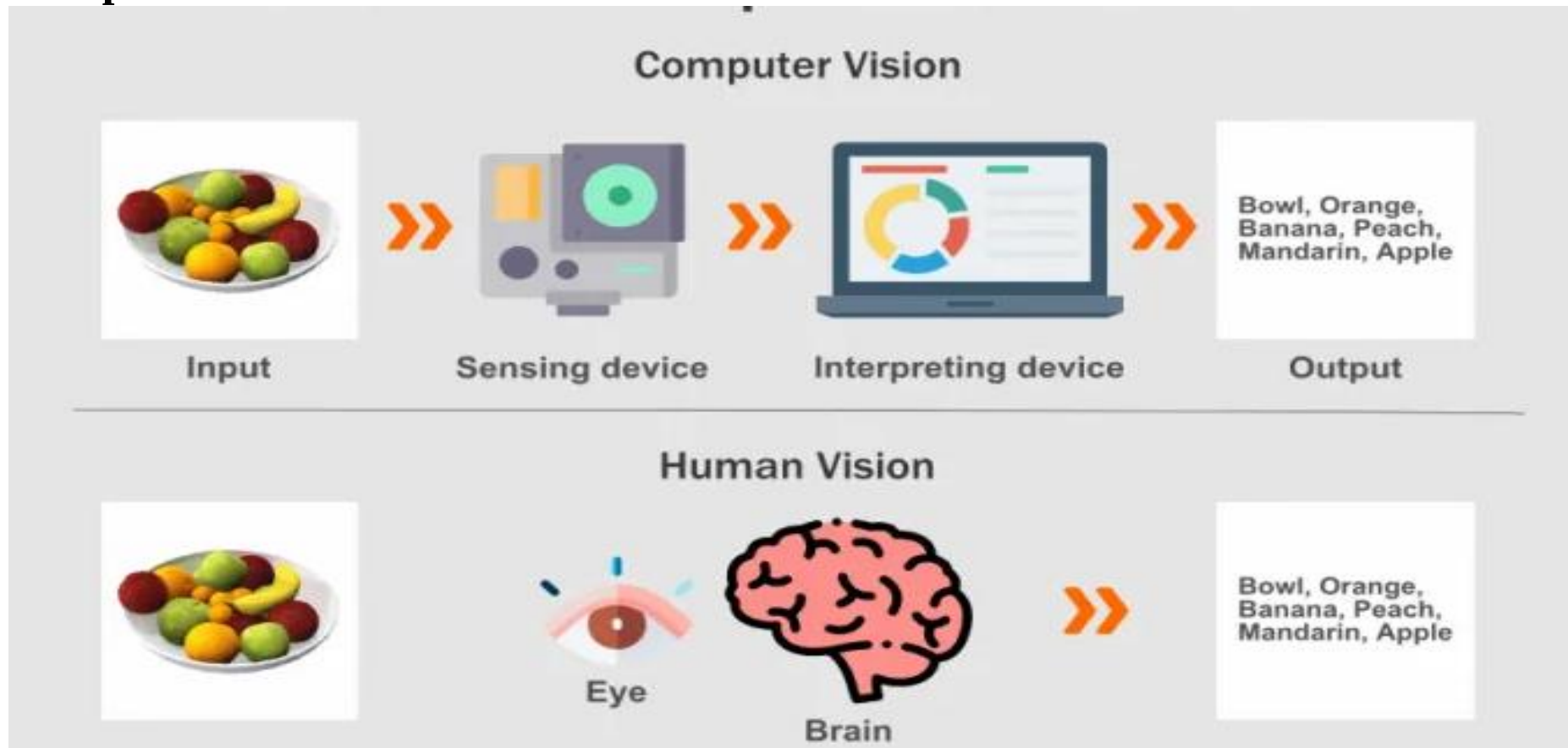


Figure 3: Computer vision and Human vision

What is computer vision?

What is in this image?

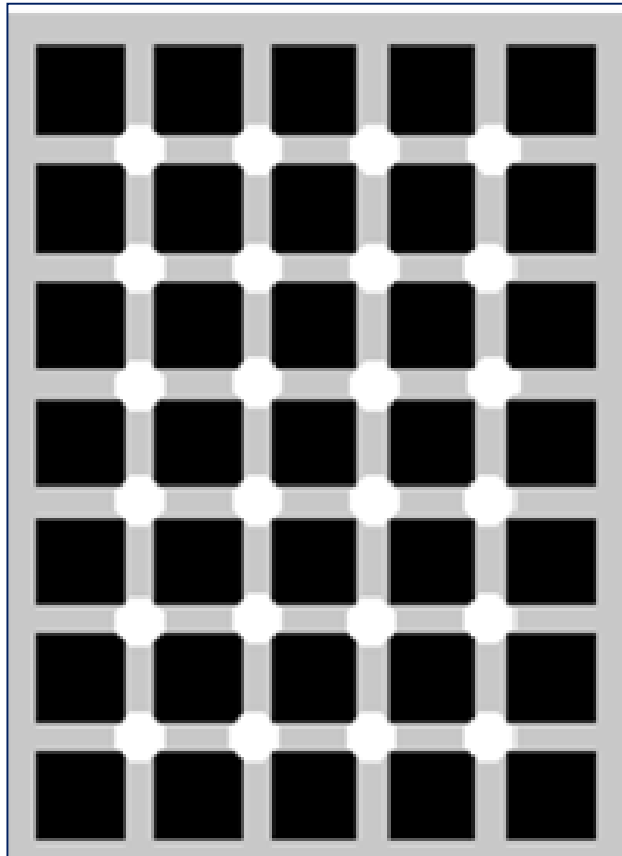
- ❖ A hand holding a man?
- ❖ A hand holding a mirrored sphere?



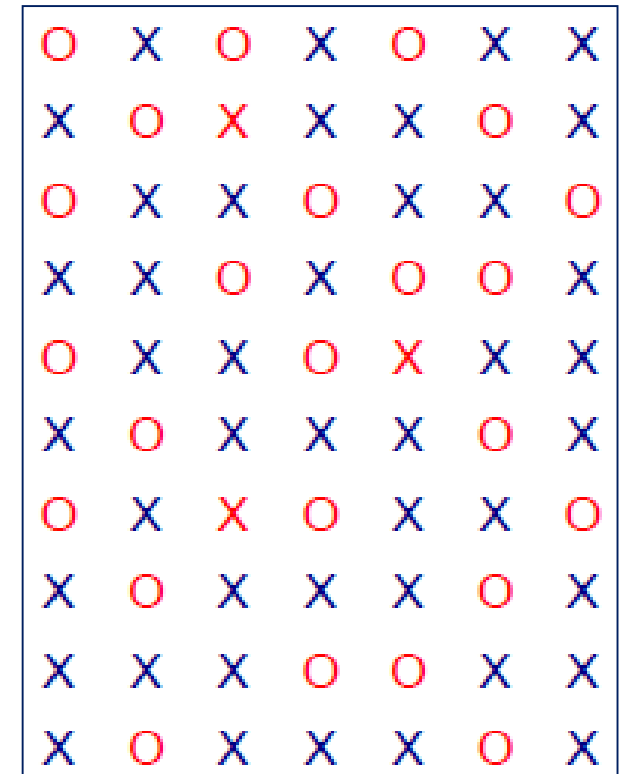
- ❖ Interpretations are ambiguous
- ❖ The forward problem (graphics)
- ❖ The “inverse problem” (vision)

Figure 4: Hand with Reflecting Sphere

What is computer vision?



Hermann Grid



How many red X's are there?

Figure 5: Optical illusion examples

What is computer vision?

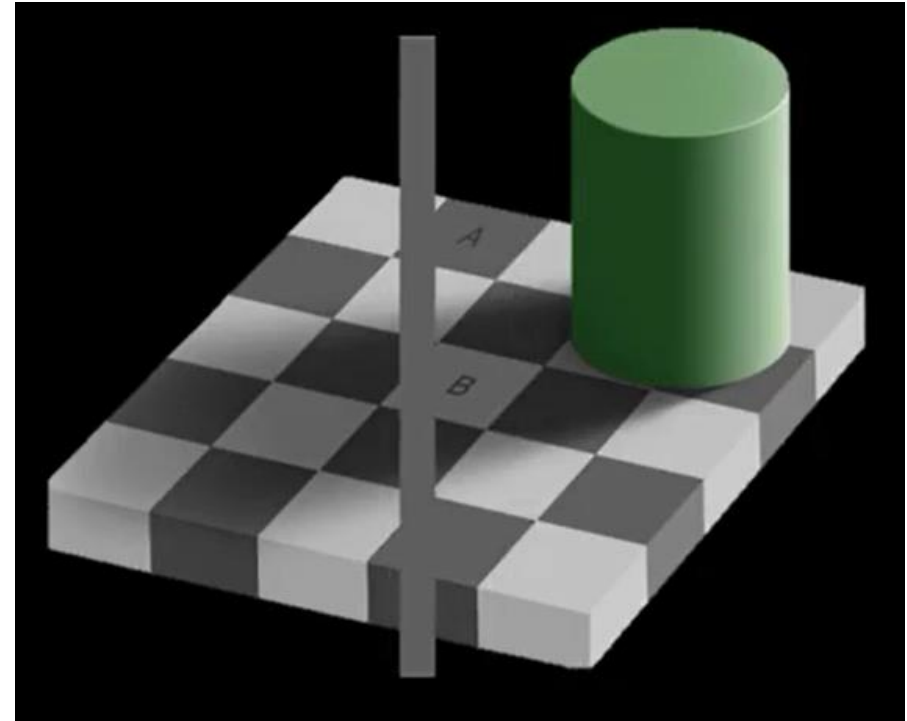
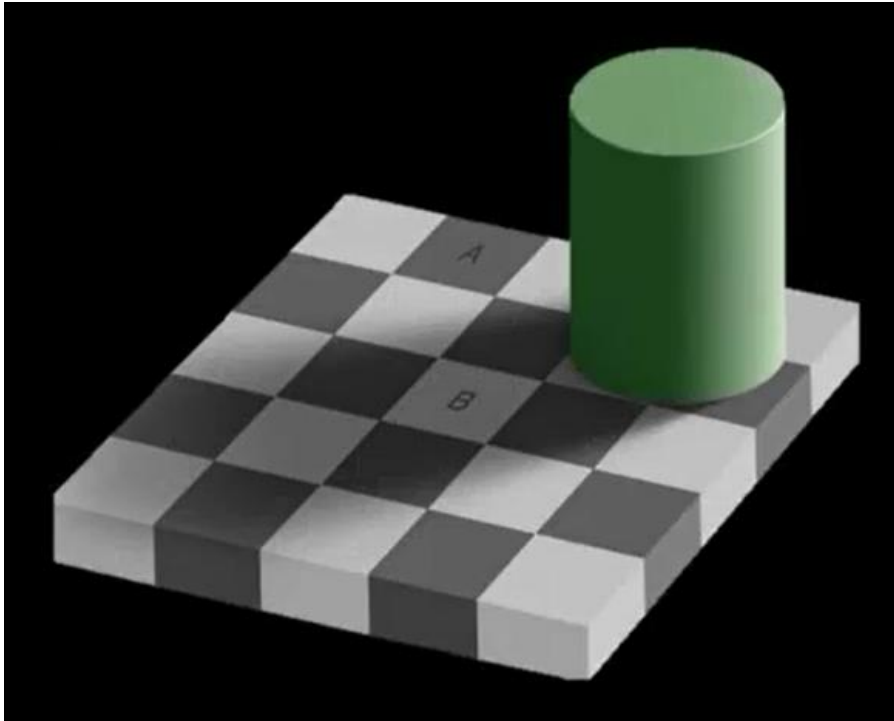


Figure 6: Visual system's attempt to discount illumination when interpreting colors

What is computer vision?

Identify the object



Figure 7: Standard room

What is computer vision?

The Challenge of making Systems Human-like

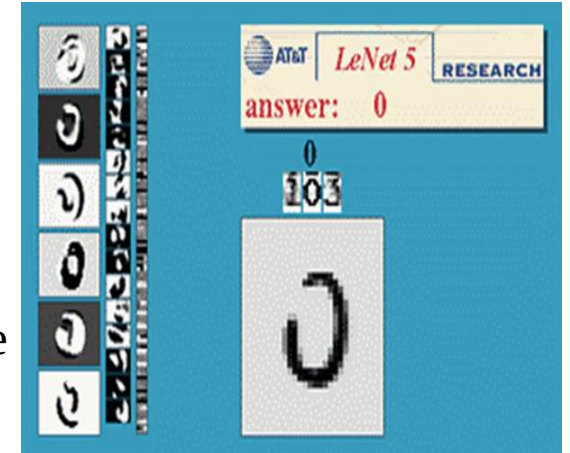
- ❖ The core problem of computer vision is object recognition.
- ❖ One reason is that we don't have a strong grasp of how human vision works.
- ❖ still learning exactly how human vision works.
- ❖ Another reason why it is such a challenging problem is because of the complexity inherent in the visual world.
- ❖ A given object may be seen from any orientation, in any lighting conditions, with any type of occlusion from other objects, and so on.

What is computer vision?

Real World Applications

❖ Optical character recognition (OCR)

- reading handwritten postal codes on letters and automatic number plate recognition (ANPR).



❖ Machine inspection

- parts inspection for quality assurance using stereo vision with specialized illumination to measure tolerances on aircraft wings, etc.



What is computer vision?

Real World Applications

❖ Retail

- Object recognition for automated checkout lanes.



❖ 3D model building

- Fully automated construction of 3D model.
E.g, aerial photographs used in systems such as Bing Maps.

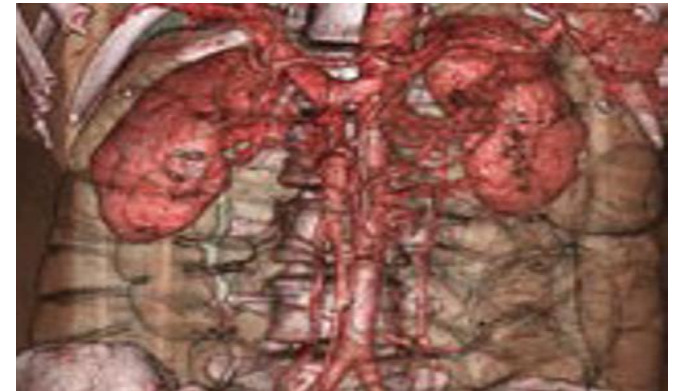


What is computer vision?

Real World Applications

❖ Medical Imaging

- To make extensive use of medical imaging data to provide us better diagnosis, treatment and predication of diseases.



❖ Automotive Safety

- detecting unexpected obstacles such as pedestrians on the street, under conditions where active vision techniques such as radar or lidar do not work well.

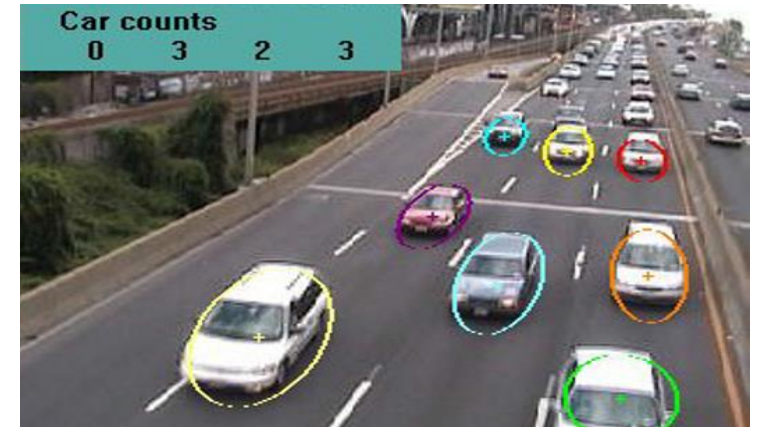


What is computer vision?

Real World Applications

❖ Surveillance

- monitoring for intruders, analyzing highway traffic, and monitoring pools for drowning victims.



❖ Fingerprint Recognition and Biometric

- for automatic access authentication as well as forensic applications.



What is computer vision?

Consumer Level Applications

❖ Stitching

- turning overlapping photos into a single seamlessly stitched panorama.



What is computer vision?

Consumer Level Applications

❖ Exposure bracketing

- merging multiple exposures taken under challenging lighting conditions (strong sunlight and shadows) into a single perfectly exposed image.



What is computer vision?

Consumer Level Applications

❖ Face Detection

- for improved camera focusing as well as more relevant image. searching.

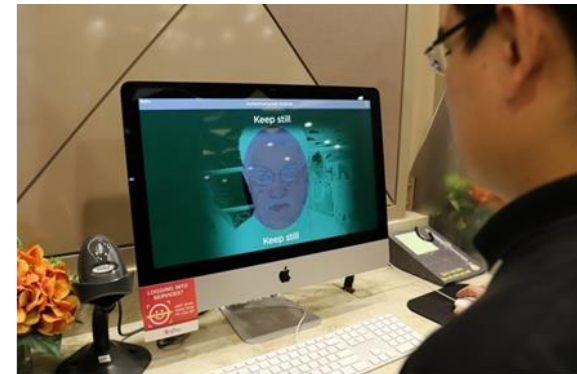
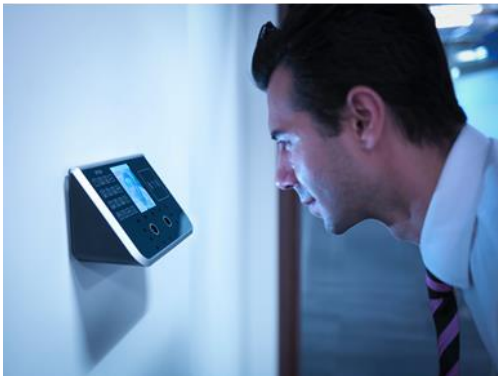


What is computer vision?

Consumer Level Applications

❖ Visual Authentication

- automatically logging family members onto your home computer as they sit down in front of the webcam.



Related Field

Computer Graphics

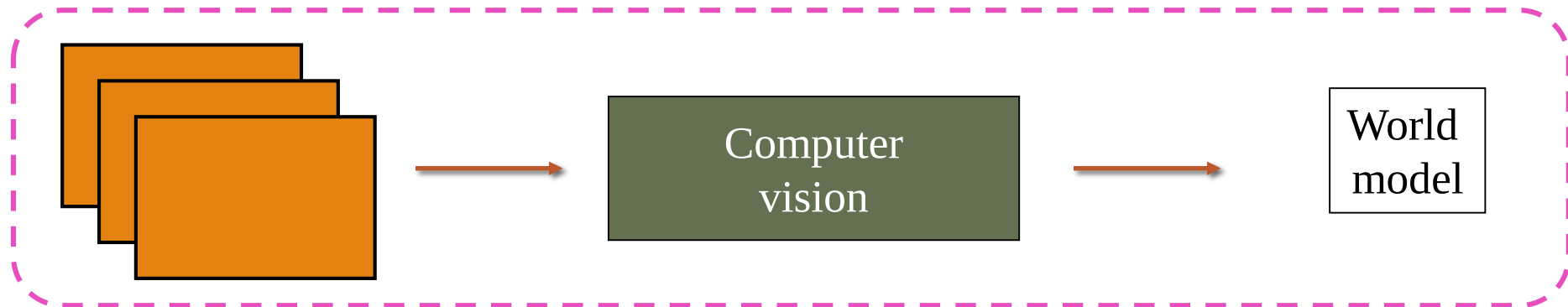
- ❖ Computer graphics is branch of computer science that deals with generating images with the aid of computers.
- ❖ “Computer graphics refers to using a computer to create or manipulate any kind of picture, image, or diagram.”, with a graphics editing program.
 - MS Paint
 - Adobe Photoshop, etc.



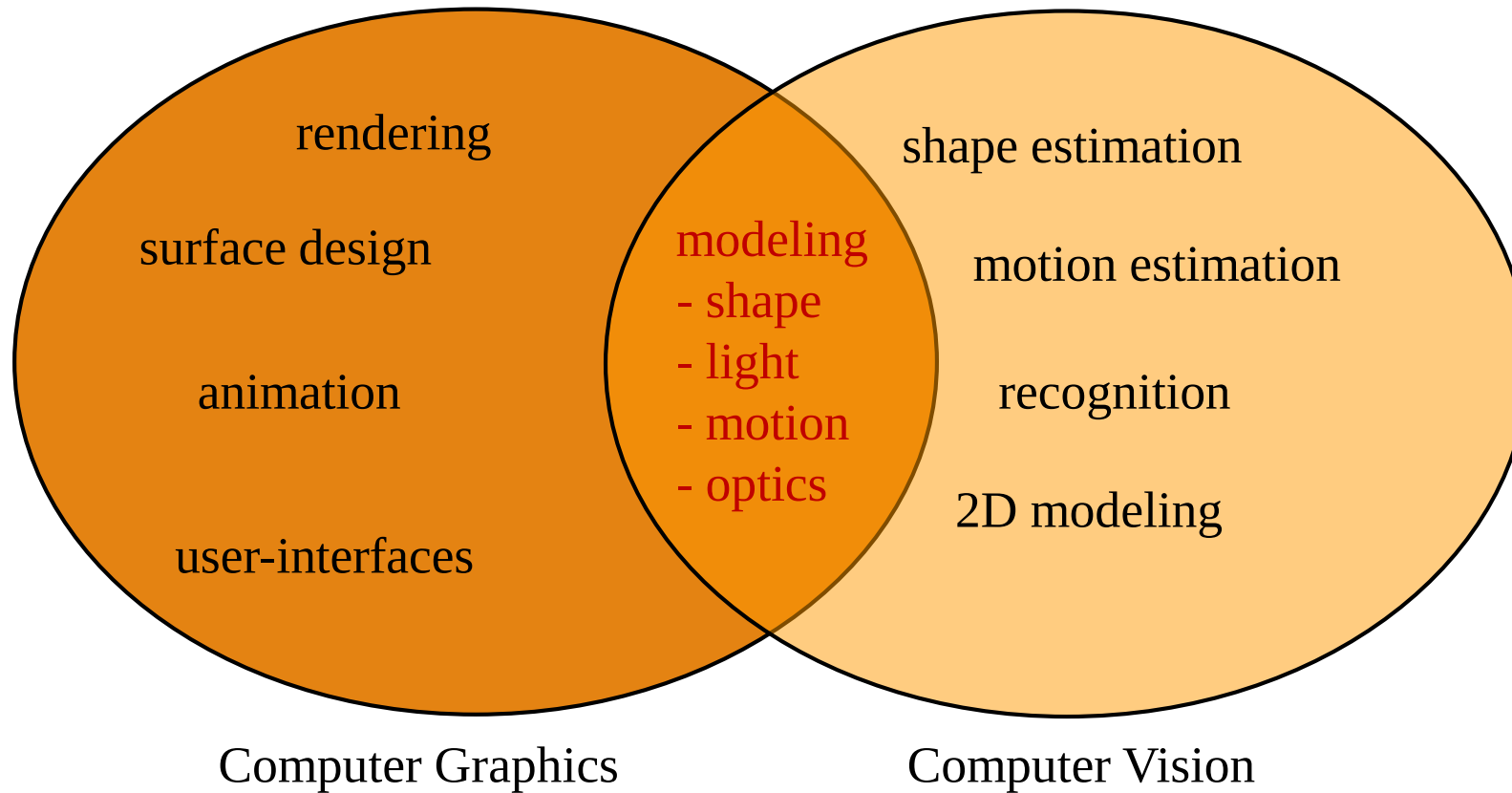
Related Field

Computer Vision

- ❖ Computer vision techniques were used in computer graphics.
 - To create image-based models of real-world objects
 - To create visual effects



Intersection of Vision and Graphics



Inverse Problem

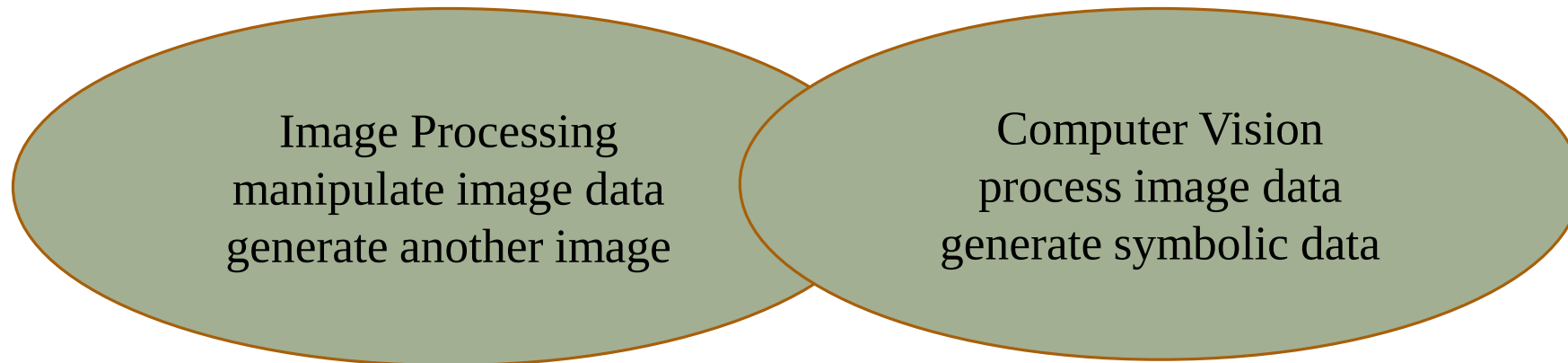
- ❖ Seek to recover some unknowns given insufficient information to fully specify the solution.
- ❖ Must resort to physics-based and probabilistic models to disambiguate between potential solutions.
- ❖ In computer vision, trying to do the inverse...
 - given one or more images, recover the scene that gave rise to these images

Forward Models

- ❖ Are developed in physics (radiometry, optics and sensor design) and in computer graphics.
- ❖ given some objects, cameras, and lighting, simulate the images that would result.
- ❖ How objects move and animate, how light reflects off their surfaces, is scattered by the atmosphere, refracted through camera lenses (or human eyes), and finally projected onto a flat (or curved) image plane.

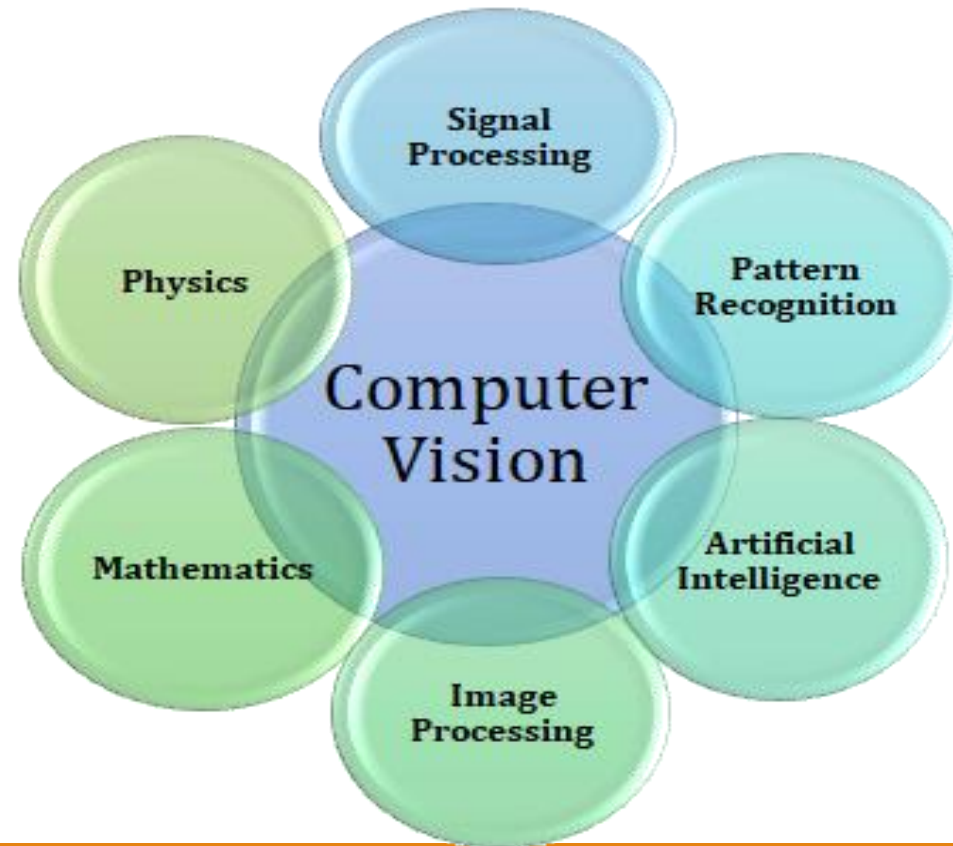
Image Processing

- ❖ Image Processing is the process of creating a new image from an existing image.
- ❖ It is a type of digital signal processing and is not concerned with understanding the content of an image.
- ❖ Examples of image processing include:
 - Normalizing, Cropping, Removing noise, etc.



Related Fields

- ❖ Computer vision combines and involves a little bit of everything in Computer Science.

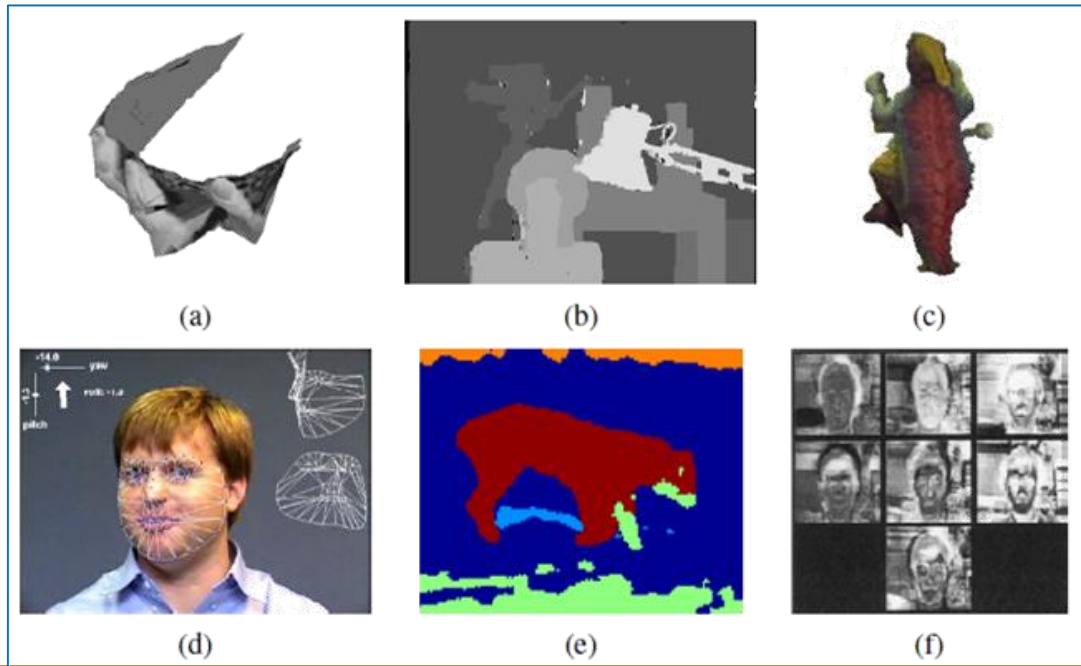


Brief History

- ❖ In the late 1960s, computer vision began at universities which were pioneering artificial intelligence.
- ❖ 1966, Marvin Minsky at MIT asked his undergraduate student Gerald Jay Sussman to “spend the summer linking a camera to a computer and getting the computer to describe what it saw”.
- ❖ In the 1970s, foundations for many of the computer vision algorithms that exist today, including extraction of edges, etc.
- ❖ In the 1980s, a lot of attention was focused on more sophisticated mathematical techniques for performing quantitative image and scene analysis.

Brief History

- ❖ In the 1990s,
 - a significant change came about with the increased interaction between the fields of computer graphics and computer vision.
 - a few of them become more active.



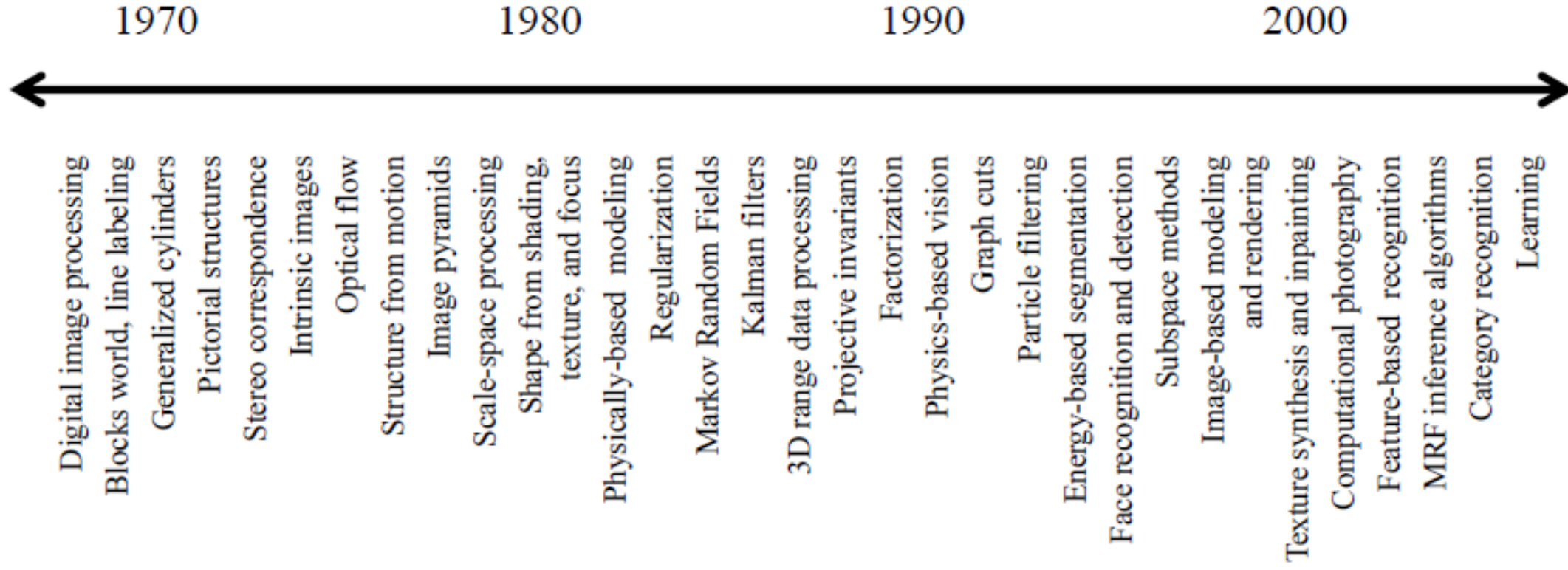
❖ Examples of computer vision algorithms from the 1990s:

- (a) factorization-based structure from motion
- (b) dense stereo matching
- (c) multi-view reconstruction
- (d) face tracking
- (e) image segmentation
- (f) face recognition

Brief History

- ❖ In the 2000s, continued to see a deepening interplay between the vision and graphics fields.
- ❖ A second notable trend during this past decade has been the emergence of feature-based techniques for object recognition.
- ❖ Recent work has seen the resurgence of feature-based methods, used in conjunction with machine learning techniques.

Brief History



Reference Books

- ❖ “Computer Vision: Algorithms and Applications” by Richard Szeliski, 1st Edition, ISBN 978-1-84882-934-3
- ❖ “Computer and Machine Vision: Theory, Algorithms, Practicalities” by E. R. DAVIES
Department of Physics Royal Holloway, 4th Edition, ISBN: 978-0-12-386908-1
- ❖ <https://www.coursera.org/>